Predicting Software Developer Stress Levels

Department of Statistics & Computer Science

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STAT 31631 – Statistical Modeling

Group Project Report – Group 02

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Group Number 02

Declaration

We do hereby declare that the work reported in this report which titled as "Predicting Software Developer Stress Levels" is originally prepared by us, Group 02 in the purpose of partial fulfilment of requirement of the Bachelor of Science Degree, Department of Statistics & Computer Science, Faculty of Science, University of Kelaniya, Sri Lanka and not for any other academic purposes. No part of this report has been submitted earlier or concurrently for the same or any other degree.

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Abstract

Software development is a demanding profession in which workload, working conditions, and individual well-being can contribute to increased stress. This study investigates factors associated with software developer stress and develops statistical models for predicting stress levels using the Developer Stress Simulation dataset. The dataset contains information on workload, sleep, technical demands, and work-environment characteristics, with Stress_Level measured on a continuous scale from 0 to 100. Exploratory data analysis was first conducted to examine variable distributions and relationships. Multiple linear regression was then used to model the continuous stress score, followed by variable selection, multicollinearity assessment, and diagnostic analysis. An ordinal logistic regression model was additionally developed by categorizing stress into ordered Low, Moderate, and High levels to provide a complementary classification perspective. The final multiple linear regression model included working hours, sleep hours, bugs, deadline days, meetings, and interruptions, and explained approximately 60.85% of the variation in stress levels. On the test data, the model achieved an R^2 of 0.5885, RMSE of 16.48, and MAE of 12.85. The ordinal logistic model identified sleep hours, working hours, deadline days, meetings, and interruptions as important predictors and achieved an overall classification accuracy of approximately 72.38%. However, the proportional-odds assumption was not fully satisfied. Overall, the findings highlight sleep, workload, meetings, and interruptions as important factors associated with software developer stress and demonstrate the usefulness of complementary statistical modeling approaches for understanding and predicting developer stress.

Keywords: Software Developer Stress, Multiple Linear Regression, Ordinal Logistic Regression, Statistical Modeling, Stress Prediction, Variable Selection, Developer Well-being

1.Introduction

1.1 Background

The software industry has become an important component of the modern digital economy, with software developers working in increasingly complex and demanding environments. Software development involves activities such as programming, debugging, testing, problem solving, collaboration, meetings, and responding to changing project requirements. Although these activities are essential for delivering software products, continuous exposure to workload and work-related pressures may negatively affect developers' well-being.

Software developer stress is therefore an important issue because prolonged or excessive stress may be associated with reduced productivity, burnout, dissatisfaction, and employee turnover. In addition to the amount of work performed, several aspects of the working environment may contribute to stress, including working hours, insufficient sleep, software bugs, deadlines, meetings, interruptions, and the complexity of software projects.

The relationship between these factors and stress is not necessarily straightforward. Different developers may experience different levels of stress under similar working conditions. Therefore, statistical methods can provide a systematic approach for identifying important factors associated with stress and for developing models capable of predicting stress levels based on observable characteristics.

In this study, statistical modeling techniques are applied to a Developer Stress Simulation dataset obtained from Kaggle. The dataset contains a continuous response variable, Stress_Level, measured on a scale from 0 to 100, together with ten potential explanatory variables representing workload, working conditions, developer experience, software complexity, and work arrangements. The project materials identify the ten predictors as Hours_Worked, Sleep_Hours, Bugs, Deadline_Days, Coffee_Cups, Meetings, Interruptions, Experience_Years, Code_Complexity, and Remote_Work.

In this study, developer stress is examined from two complementary perspectives. First, the continuous stress score is modeled using multiple linear regression to identify factors associated with changes in the overall stress level. Second, the continuous stress score is converted into ordered stress categories representing different levels of stress, and ordinal logistic regression is used to investigate the likelihood of developers belonging to higher stress categories. Using both approaches provides a more comprehensive assessment of developer stress by considering both the magnitude of the stress score and its severity category.

1.2Research Problem

Software developers frequently work under conditions involving deadlines, varying workloads, technical problems, meetings, interruptions, and different levels of software complexity. These factors can potentially influence the level of stress experienced by developers. However, identifying which factors provide meaningful information about stress and determining whether these factors can be combined into a useful statistical prediction model requires systematic analysis.

The central problem addressed by this study is therefore:

- How can statistical modeling be used to identify important factors associated with software developer stress and predict developer stress levels using workload, behavioral, and work-environment characteristics?

The project uses a dataset containing 526 observations, which was reduced to 506 observations after data cleaning. Stress_Level is treated as the continuous response variable, while ten variables are considered as potential predictors.

Therefore, there is a need to identify the factors associated with software developer stress and to develop statistical models that can provide useful predictions. In particular, it is useful to examine stress both as a continuous measure and as an ordered severity outcome. This study therefore applies multiple linear regression to predict the continuous stress score and ordinal logistic regression to investigate the probability of belonging to higher stress categories.

1.3 Research Questions

Based on the research problem, the study addresses the following questions:

Q1. What are the main characteristics and distribution of software developer stress levels in the selected dataset?

Q2. What relationships exist between software developer stress levels and the available workload, behavioral, and work-environment variables?

Q3. Which predictor variables provide statistically significant information for explaining software developer stress levels?

Q5. Can variable selection produce a more appropriate and interpretable regression model while maintaining satisfactory predictive performance?

Q6. To what extent does the final selected regression model satisfy the assumptions required for reliable statistical inference?

Q7. How accurately can ordinal logistic regression classify developers into Low, Moderate, and High stress categories?

Q8. Does the ordinal logistic regression model satisfy the proportional-odds assumption?

1.4 Objectives

1.4.1 Main Objective

To develop and compare statistical models for identifying important factors and predicting software developer stress levels using continuous and ordinal modeling approaches.**1.4.2 Specific Objectives**

The study aims to:

1. To explore the distribution and characteristics of software developer stress levels.
2. To examine relationships between stress level and workload, behavioral, technical, and work-environment variables.
3. To develop a multiple linear regression model for predicting the continuous Stress_Level.

4. To identify important predictors and obtain a parsimonious multiple linear regression model through variable selection.
5. To assess multicollinearity and the assumptions of the multiple linear regression model.
6. To develop an ordinal logistic regression model using ordered stress categories.
7. To identify predictors associated with higher stress categories using odds ratios.
8. To assess the proportional-odds assumption of the ordinal logistic regression model.
9. To evaluate and compare the predictive performance of the statistical models using appropriate measures.

These objectives directly support the project requirements to define a clear main objective and a set of specific objectives that break the overall research goal into manageable components.

1.5 Scope of the Study

This study focuses on predicting software developer stress levels using the variables available in the Developer Stress Simulation dataset.

The response variable is Stress_Level, a continuous score ranging from 0 to 100. For descriptive interpretation, the project materials classify scores of 0–33 as low stress, 34–66 as moderate stress, and 67–100 as high stress.

The explanatory variables considered in the study are:

- Hours_Worked
- Sleep_Hours
- Bugs
- Deadline_Days
- Coffee_Cups
- Meetings
- Interruptions
- Experience_Years
- Code_Complexity
- Remote_Work

The analysis focuses on statistical relationships and predictive modeling rather than clinical diagnosis of psychological or medical conditions. The study therefore interprets the model in terms of statistical associations and predictions within the available dataset.

The analysis follows a structured statistical modeling process consisting of data preprocessing, exploratory data analysis, construction of an initial regression model, diagnostic assessment, variable selection, further model development, and evaluation of the resulting models. The project's current workflow records the progression from the full model through diagnostic checking and stepwise model selection.

02. Literature Review

2.1 Introduction

Software development is a demanding, knowledge-intensive occupation involving problem solving, concentration, collaboration, and changing technical requirements. These demands can affect developers' well-being and contribute to workplace stress.

Previous studies have examined developer and IT-professional stress in relation to factors such as workload, interruptions, and sleep using different research and statistical approaches. The studies considered in this project provide a foundation for understanding these factors while also highlighting limitations in existing approaches.

This literature review therefore examines key factors associated with software developer stress, relevant statistical modeling approaches, and the research gap addressed by the present study.

2.2 Software Developer Well-being and Stress

Software development is a knowledge-intensive occupation involving continuous problem solving, concentration, collaboration, and adaptation to changing requirements. These demands can affect developers' psychological well-being and contribute to workplace stress.

Graziotin et al. (2018) investigated well-being in software development and highlighted the importance of developers' emotional experiences. Their findings, as summarized in the project literature review, indicate that work interruptions can contribute to increased stress. However, the study used qualitative interviews with a relatively small sample, limiting generalizability.

This motivates that in our study to examine interruptions quantitatively together with other potential stress predictors.

2.3 Workload and Burnout

Workload is an important potential source of occupational stress. In software development, workload can be reflected through working hours, deadlines, meetings, and the volume of technical tasks.

Fagerlin et al. (2020) examined the relationship between workload and burnout using survey data and ANOVA. Their findings indicated an association between workload and burnout. However, the study's approach provides limited consideration of multiple predictors simultaneously.

The present study addresses this by considering several workload-related variables, including Hours_Worked, Deadline_Days, Meetings, and Interruptions, within a multiple regression framework.

2.4 Sleep and Developer Stress

Sleep is an important aspect of recovery and well-being, and insufficient sleep may reduce an individual's ability to manage work-related demands.

Sinclair et al. (2021) investigated stress among IT professionals using logistic regression, with the findings summarized in the project literature review indicating that sleep was a useful predictor of stress.

In the present study, Sleep_Hours is included as a potential predictor of the continuous Stress_Level response, allowing its relationship with stress to be examined alongside other workplace factors.

2.5 Work Interruptions and Meetings

Interruptions can disrupt periods of concentrated technical work and may increase the cognitive demands placed on developers. Previous research identified by the group suggests that interruptions are associated with increased developer stress.

Therefore, the present study includes both **Interruptions** and **Meetings** as potential predictors, allowing their contribution to stress to be assessed while controlling for other variables.

2.6 Statistical Modeling of Stress

2.6.1 Multiple Linear Regression

Statistical modeling provides a systematic approach for examining relationships between a response variable and multiple explanatory variables. Since Stress_Level is a continuous variable, multiple linear regression is used as the primary modeling approach.

The statistical modeling principles discussed by Hastie, Tibshirani, and Friedman (2009) provide a foundation for regression and model-selection techniques used in this study.

Multiple linear regression allows the relationship between each predictor and stress to be examined while controlling for the other predictors in the model. The general form is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

where Yrepresents Stress_Level, $X_1, \dots, X_p$represent the explanatory variables, and ϵ represents the random error.

2.6.2 Ordinal Logistic Regression

Ordinal logistic regression is appropriate when the response variable consists of ordered categories. Unlike multiple linear regression, which models a continuous response, ordinal logistic regression models the probability of an observation falling into a higher or lower category while preserving the natural ordering of the outcome categories. In the present study, the continuous stress score is additionally represented using ordered stress categories, allowing the factors associated with higher stress severity to be investigated.

2.8 Ordinal Logistic Regression and Proportional Odds

Ordinal logistic regression commonly uses a proportional-odds formulation, which assumes that the relationship between each predictor and the cumulative odds of the ordered outcome remains consistent across the different category thresholds. This assumption should be assessed before interpreting an ordinal logistic model. In this study, the proportional-odds assumption is evaluated after fitting the ordinal regression model, and further threshold-specific analysis is considered if the assumption is violated.

2.7 Variable Selection and Model Parsimony

When several potential predictors are available, including unnecessary variables can make a model more complex and difficult to interpret. Variable selection can therefore be used to identify a suitable subset of predictors.

The STAT 31631 practical guide discusses best subset selection, forward selection, backward elimination, and stepwise regression as approaches for selecting useful predictor combinations.

In this study, variable-selection procedures are used to investigate whether a more parsimonious model can provide satisfactory explanatory and predictive performance. Model comparison considers appropriate criteria such as adjusted R^2, AIC, BIC, and prediction error where applicable.

2.8 Multicollinearity in Regression Models

Multicollinearity occurs when explanatory variables are strongly related to one another. It can make coefficient estimates unstable and make it difficult to determine the individual contribution of correlated predictors.

The present study includes correlation and VIF analysis as part of the regression modeling process to ensure that relationships among predictors do not adversely affect the reliability and interpretation of the fitted models.

2.10 Research Gap

The reviewed studies provide useful evidence regarding factors associated with software developer and IT-professional stress. However, the approaches differ substantially.

The literature comparison prepared by the group summarizes the main differences as follows:

Study	Method	Main Finding	Identified Limitation
Fagerlin et al. (2020)	Survey + ANOVA	Workload linked to burnout	Single method
Graziotin et al. (2018)	Qualitative interviews	Interruptions associated with stress	Small sample
Sinclair et al. (2021)	Logistic regression	Sleep useful for predicting stress	No regularization
Present study	Multiple Linear Regression	Comparative statistical modeling of stress	Dataset-specific analysis

These differences highlight a gap in applying a single comparative statistical framework containing multiple workload, behavioral, technical, and work-environment variables to predict a continuous measure of developer stress.

The present study addresses this gap by considering multiple candidate predictors simultaneously and following a structured modeling process involving exploratory analysis, multiple linear regression, multicollinearity assessment, variable selection, model diagnostics, and predictive evaluation.

2.12 Summary of Literature Review

Previous research indicates that software developer and IT-professional well-being can be influenced by factors related to workload, interruptions, and sleep. The reviewed studies also demonstrate that different statistical and research approaches can be used to investigate these relationships.

However, the existing approaches identified in the project's literature review differ in terms of methodology, sample characteristics, and variables considered. The present study therefore adopts a multiple regression framework to examine several potential predictors simultaneously.

The literature review provides the theoretical and methodological foundation for the subsequent analysis. In particular, it motivates the inclusion of workload, sleep, technical, and work-environment variables and supports the use of regression modeling, variable selection, multicollinearity assessment, and model diagnostics.

03.Methodology

3.1 Research Design

This study follows a quantitative statistical modeling approach to investigate factors associated with software developer stress levels and to develop predictive regression models. The analysis consists of data preprocessing, exploratory data analysis, multiple linear regression, multicollinearity assessment, variable selection, model diagnostics, and model evaluation.

The overall modeling process begins with the preparation and exploration of the dataset, followed by fitting an initial model containing the available predictors. Diagnostic procedures are then used to assess the adequacy of the model and identify potential issues. Variable-selection techniques are subsequently applied to obtain a more suitable and interpretable model.

Two complementary statistical modeling approaches were used: multiple linear regression for the continuous stress score and ordinal logistic regression for ordered stress categories.

A full model is fitted first and then assessed using residual analysis, multicollinearity checks, and model-selection procedures.

3.2 Data Source and Dataset

The study uses the 'Developer Stress Simulation' dataset obtained from Kaggle. The dataset contains information related to software developers' workload, working conditions, behavioral characteristics, and technical working environment.

The original dataset contained 526 observations. After the data-cleaning procedures described below, 506 observations remained for the statistical modeling analysis. The response variable is 'Stress_Level', which is a continuous measure ranging from 0 to 100.

The project also uses a categorical interpretation of the stress score for descriptive purposes:

- **0–33:** Low stress
- **34–66:** Moderate stress
- **67–100:** High stress

The primary statistical analysis, however, uses the continuous Stress_Level variable as the response.

3.3 Variables Used in the Study

The study considers Stress_Level and Stress_Category as the response variables in MLR and Ordinal Logistic regression accordingly and ten variables as potential predictors.

Variable	Type	Description
Stress_Level	Continuous	Overall software developer stress score
Stress_Category	Ordered	Low, Moderate, High stress categories
Hours_Worked	Quantitative	Number of hours worked
Sleep_Hours	Quantitative	Hours of sleep
Bugs	Quantitative	Number of software bugs
Deadline_Days	Quantitative	Deadline-related work measure
Coffee_Cups	Quantitative	Number of coffee cups consumed
Meetings	Quantitative	Number of meetings
Interruptions	Quantitative	Number of work interruptions
Experience_Years	Categorical/Ordinal	Developer experience level
Code_Complexity	Categorical/Ordinal	Code complexity level
Remote_Work	Categorical/Ordinal	Remote-work level/status

The R analysis separately examines the categorical variables and their observed levels before model fitting to ensure that they are appropriately represented in the modeling dataset.

3.4 Data Preprocessing

Data preprocessing was performed before statistical modeling to ensure that the dataset was suitable for analysis. The preprocessing consisted of several stages.

3.4.1 Data Import and Initial Inspection

The dataset was imported into RStudio using the read.csv() function. Initial inspection was conducted using functions such as head(), dim(), and names() to examine the observations, dimensions, and variable names.

This initial inspection was performed to understand the structure of the raw dataset before applying any modifications.

3.4.2 Identification of Unnecessary Columns

The dataset was examined for columns containing no observations. The number of non-missing observations in each column was calculated before removing unnecessary columns.

Columns containing zero non-missing observations were removed because they provide no information for statistical analysis. Importantly, only completely empty columns were removed at this stage, while the original raw dataset was retained separately.

3.4.3 Investigation of Missing Values

Missing values were investigated both across observations and variables. Complete-case analysis was initially used to identify observations containing missing information, followed by examination of the number of missing values in each variable.

Particular attention was given to missing values in the response variable, Stress_Level, because a response value is required for supervised regression modeling.

3.4.4 Handling Missing Response Values

Observations with missing Stress_Level values were excluded from the final modeling dataset. Since the response variable is required to estimate a regression model, observations without a stress score cannot contribute to model fitting or evaluation.

After this preprocessing stage, the modeling dataset contained 506 observations.

3.4.5 Examination of Categorical Variables

The structure of the modeling dataset was examined using R functions such as str() and summary() to distinguish numerical and categorical variables.

Frequency tables were also produced for Experience_Years, Code_Complexity, and Remote_Work. Unique values were inspected to identify blank or unexpected categories before these variables were considered in the regression analysis.

3.4.6 Construction of the Ordinal Stress Outcome

For the ordinal logistic regression analysis, the continuous Stress_Level variable was transformed into an ordered categorical response, Stress_Category. The categories were arranged from lower to higher stress, allowing the ordinal nature of the response to be incorporated into the model.

The category boundaries were determined using the training data to avoid using information from the test set during model construction. The resulting categories were ordered as Low < Moderate < High.

- Low: Stress_Level ≤ 64.08702
- Moderate: 64.08702 < Stress_Level < 100
- High: Stress_Level = 100

3.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the response and explanatory variables and to identify potential relationships before fitting regression models.

Descriptive statistics were used to summarize quantitative variables, while frequency distributions were used to examine categorical variables. Graphical techniques were also used to investigate the distribution of Stress_Level and relationships between the response and potential predictors.

The main EDA techniques used in the study include:

- descriptive statistics;
- frequency distributions;
- histograms;
- boxplots;
- scatter plots;
- correlation analysis; and
- graphical comparisons between categorical predictors and stress levels.

In addition to examining the continuous stress score, the distribution of the resulting ordered stress categories was examined before fitting the ordinal logistic regression model.

3.6 Multiple Linear Regression

Because Stress_Level is a continuous response variable, Multiple Linear Regression (MLR) was selected as the primary statistical modeling technique.

multiple regression is a method for modeling the relationship between a response variable and several explanatory variables and emphasize evaluating coefficients, model significance, R^2, adjusted R^2, and residual behavior.

The general multiple linear regression model is expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

where:

- $Y$is the response variable, Stress_Level;
- $X_1, \ldots, X_p$are the predictor variables;
- $\beta_0$is the intercept;
- $\beta_1, \ldots, \beta_p$are regression coefficients; and
- ϵ represents the random error.

Initially, a **full regression model** containing all candidate predictors was fitted. This provides a baseline model from which subsequent diagnostic and variable-selection procedures can be performed.

3.7 Multicollinearity Assessment

Multicollinearity was assessed to determine whether strong relationships existed among the explanatory variables.

Two main approaches were considered:

Correlation Analysis

A correlation matrix was used to examine pairwise relationships between quantitative predictors.

Variance Inflation Factor

The Variance Inflation Factor (VIF) was used to quantify the degree of multicollinearity among predictors.

VIF is an important diagnostic for multicollinearity, with larger VIF values indicating greater concern.

3.8 Variable Selection

After fitting and assessing the full regression model, variable-selection procedures were applied to investigate whether a smaller set of predictors could provide a more suitable model.

Variable selection is important when several potential predictors are available because unnecessary variables can increase model complexity and reduce interpretability.

In this study, a stepwise model-selection approach was used as part of the model-building process. Candidate models were assessed using statistical and model-performance criteria, while diagnostic checks were performed between modeling stages.

The selection process followed the general principle of:

Full model → model assessment → variable selection → diagnostic assessment → refined model

This iterative approach of model-selection procedures should be followed by further checks of residuals, functional form, outliers, and influential observations.

3.9 Ordinal Logistic Regression

3.9.1 Model Specification

Ordinal logistic regression was used to model the ordered stress categories. The model estimates cumulative odds of being in a higher stress category while accounting for the predictor variables.

The cumulative-logit form:

$$\log\left[\frac{P(Y \leq j)}{P(Y > j)}\right] = \alpha_j - \beta_1 X_1 - \cdots - \beta_p X_p$$

where j represents the cumulative category thresholds.

3.9.2 Odds Ratios

Regression coefficients were exponentiated to obtain odds ratios, allowing the effects of predictors to be interpreted in terms of changes in the odds of being in a higher stress category.

3.9.3 Proportional-Odds Assumption

The proportional-odds assumption was assessed using the Brant test. This assumption requires the relationship between the predictors and the cumulative odds to remain consistent across the different thresholds of the ordinal response.

3.9.4 Ordinal Model Evaluation

- Confusion matrix
- Accuracy
- Precision
- Recall
- F1-score

3.10 Model Evaluation and Comparison

Model	Response	Evaluation
MLR	Continuous Stress_Level	R^2, Adjusted R^2, RMSE, MAE
Ordinal Logistic	Stress_Category	Accuracy, Precision, Recall, F1, Confusion Matrix, AIC

3.11 Model Diagnostics

Model diagnostics were performed to assess whether the fitted regression models satisfied the assumptions required for reliable statistical inference and prediction.

The following diagnostic aspects were considered:

3.11.1 MLR Diagnostics

Linearity

Residual-versus-fitted plots were examined to determine whether the relationship between the response and predictors was appropriately represented by the linear model.

Homoscedasticity

Residual plots were examined for evidence of non-constant variance. A suitable regression model should have residual variability that is approximately constant across fitted values.

Normality of Residuals

Normal Q-Q plots were used to assess whether the residuals were approximately normally distributed.

3.11.2 Ordinal Logistic Diagnostics

The ordinal logistic regression model was assessed using the proportional-odds assumption. The Brant test was used to determine whether the effects of the predictors remained consistent across the different stress-category thresholds. If the assumption was violated, threshold-specific binary logistic regression models were examined to further investigate the predictors contributing to the violation.

3.11.3 Independence

Residual patterns were examined to identify potential systematic patterns that could indicate dependence among observations.

3.11.4 Influential Observations

Diagnostic measures were considered to identify observations that could have an unusually large influence on the fitted regression model.

These diagnostic procedures follow the regression and model-building principles, where residual-versus-fitted plots, Q-Q plots, and residual analysis are used to assess model assumptions.

3.12 Model Evaluation

The candidate models were compared using appropriate measures of explanatory and predictive performance.

The main measures considered include:

R^2

Measures the proportion of variation in Stress_Level explained by the model.

Adjusted R^2

Provides an adjusted measure of explanatory power that accounts for the number of predictors included in the model.

RMSE

The Root Mean Squared Error measures the typical magnitude of prediction errors, with lower values indicating better predictive performance.

AIC

The Akaike Information Criterion considers both model fit and model complexity, with lower values generally preferred when comparing candidate models.

Where applicable, **cross-validation** was also considered to assess predictive performance beyond the data used for model fitting. The project objectives specifically identify R^2, RMSE, and cross-validation as measures for evaluating model accuracy.

3.13 Statistical Software

All data preprocessing, exploratory analysis, statistical modeling, diagnostics, and model evaluation were performed using **R/RStudio**.

R was selected because it provides functions and packages required for regression analysis, statistical diagnostics, variable selection, and graphical analysis.

The analysis was documented using reproducible R code, allowing the statistical procedures and results to be replicated.

04.Results and Discussion

4.1 Introduction

This chapter presents the results obtained from the analysis of software developer stress levels. The analysis was conducted using exploratory data analysis, multiple linear regression, and ordinal logistic regression. The continuous Stress_Level was first examined to understand its distribution and relationships with the explanatory variables. Multiple linear regression was then used to model the continuous stress score, while ordinal logistic regression was used as a complementary approach to classify developers into ordered stress categories. Model diagnostics and predictive performance were also evaluated.

4.2 Exploratory Data Analysis

4.2.1 Descriptive Statistics

Descriptive statistics were calculated for the continuous variables to summarize their central tendency and variability.

Table 4.1: Descriptive Statistics of Numerical Variables

Variable	Mean	Min	Q1	Median	Q3	Max
Hours_Worked	**9.445**	**4.000**	**6.000**	**9.000**	**12.750**	**16.000**
Sleep_Hours	**5.467**	**3.000**	**4.000**	**5.000**	**7.000**	**10.000**
Bugs	**24.810**	**0.000**	**12.000**	**25.000**	**37.000**	**50.000**
Deadline_Days	**31.110**	**0.000**	**16.000**	**31.000**	**45.000**	**60.000**
Coffee_Cups	**4.888**	**0.000**	**2.000**	**5.000**	**7.750**	**10.000**
Meetings	**10.060**	**0.000**	**4.000**	**10.000**	**16.000**	**20.000**
Interruptions	**4.919**	**0.000**	**2.000**	**5.000**	**7.000**	**10.000**
Stress_Level	**75.701**	**0.493**	**55.828**	**87.741**	**100.000**	

The descriptive results show substantial variation among the workload and behavioral variables. The mean stress level was 75.701, while the median was 87.741, indicating that a large proportion of observations were concentrated toward the higher end of the stress scale.

4.2.2 Distribution of Numerical Variables

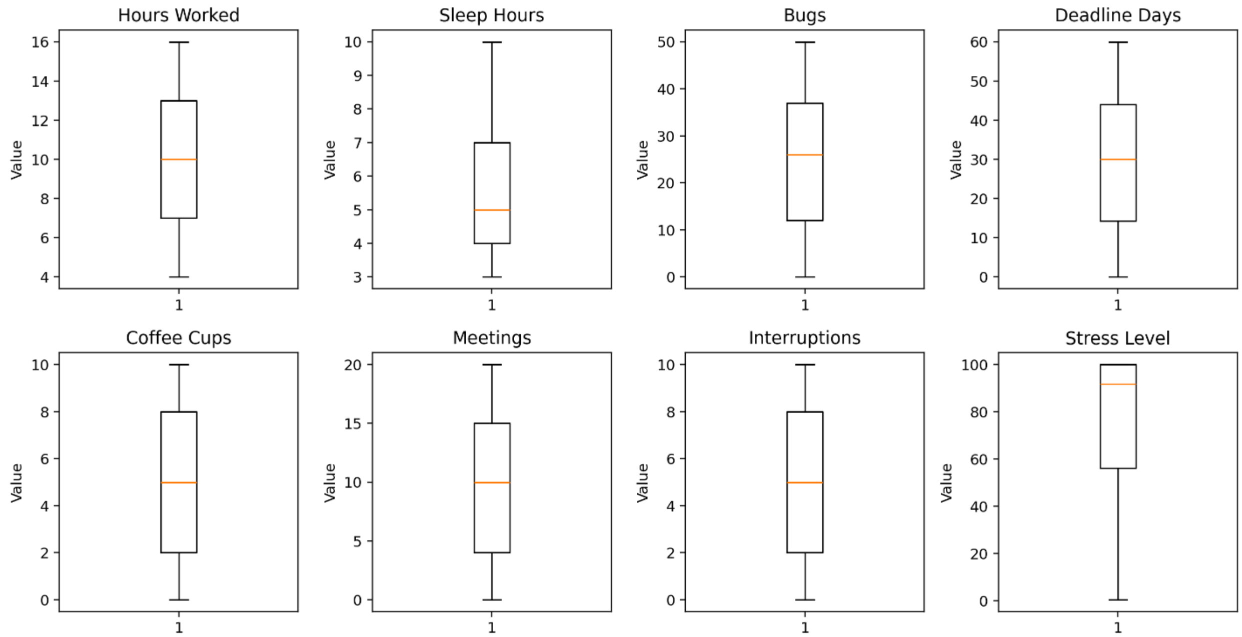


Figure 4.1: Boxplots of Numerical Variables

The boxplots provide an overview of the distributions and variability of the numerical variables. Stress_Level shows a relatively high central tendency compared with many of the explanatory variables. The boxplots also indicate the presence of observations with relatively extreme values, particularly for the stress response.

4.2.3 Distribution of Stress Level

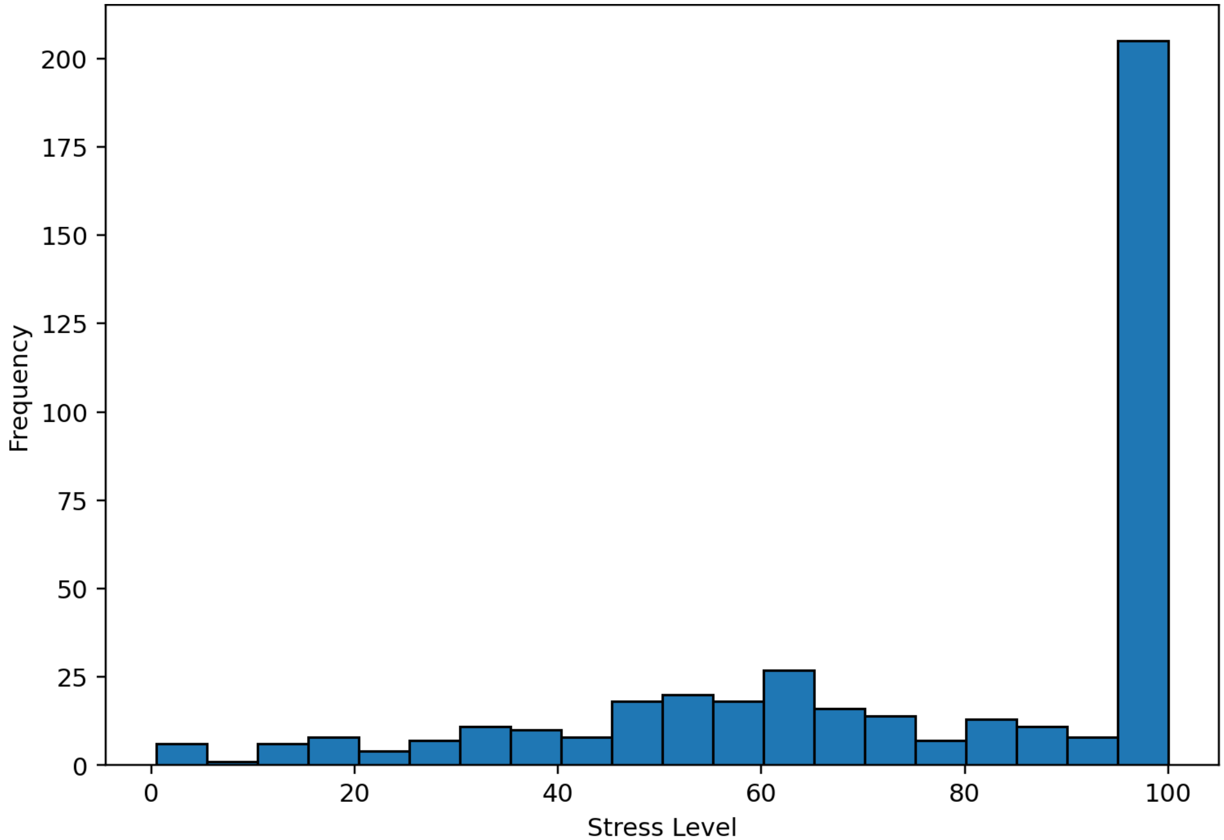


Figure 4.2: Distribution of Stress Level

The histogram shows that stress levels are concentrated toward the higher end of the scale, with a substantial number of observations at or near 100. This concentration is important when interpreting the regression results because it indicates that the response variable is not uniformly distributed across its 0–100 range.

4.2.4 Correlation Analysis

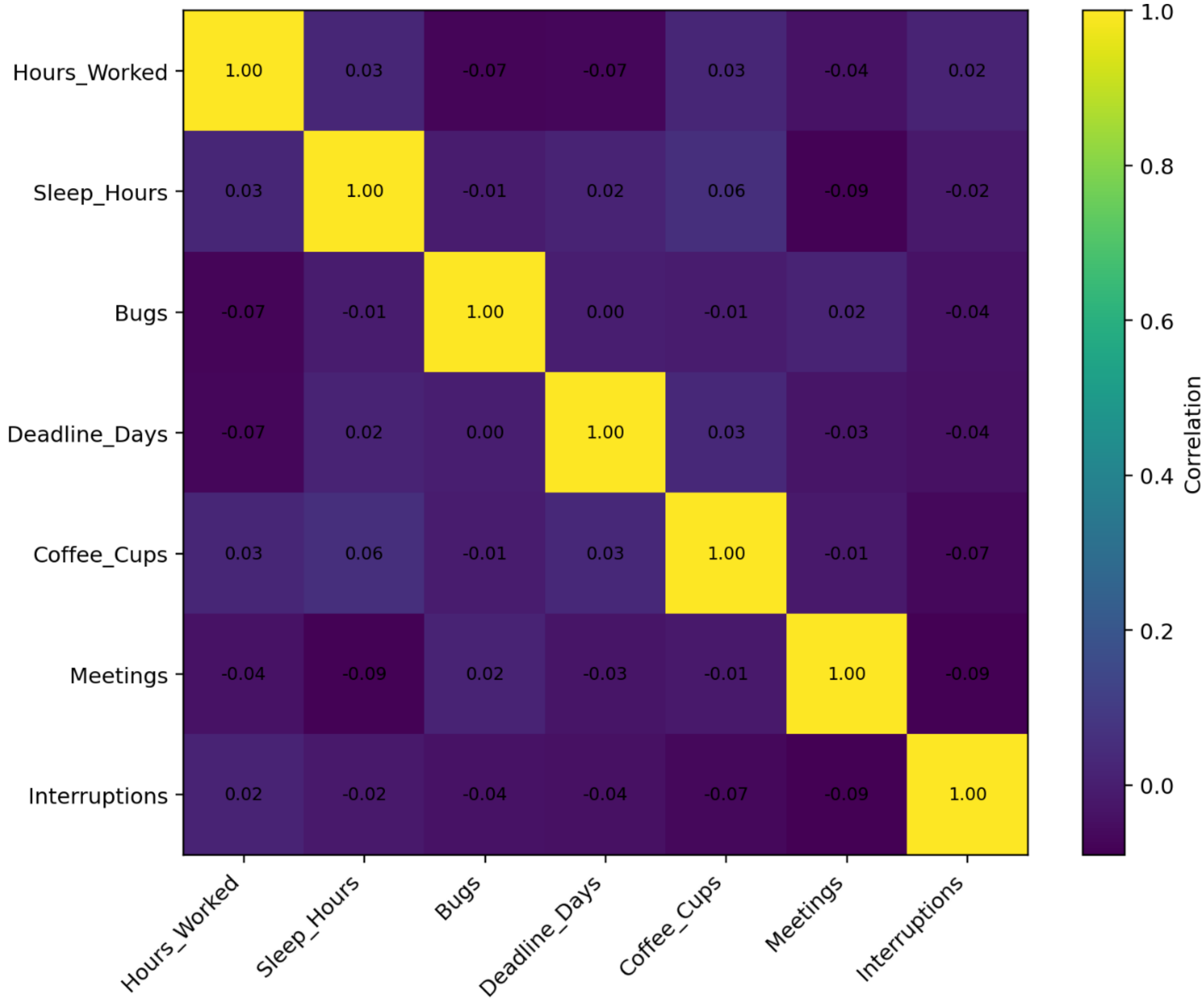


Figure 4.3: Correlation Matrix of Numerical Predictors

The correlation analysis was used to examine the relationships among the numerical predictors before fitting the regression models. The results indicate that the predictors do not exhibit extremely strong pairwise correlations. This provides an initial indication that severe multicollinearity is unlikely to be a major issue.

The relationship between Sleep_Hours and Stress_Level was particularly notable, with a relatively strong negative association. Hours_Worked and Meetings showed positive associations with stress.

4.3 Multiple Linear Regression Analysis

4.3.1 Final Multiple Linear Regression Model

Following model selection, the final multiple linear regression model included six predictors:

- Hours_Worked
- Sleep_Hours
- Bugs
- Deadline_Days
- Meetings
- Interruptions

The estimated model was:

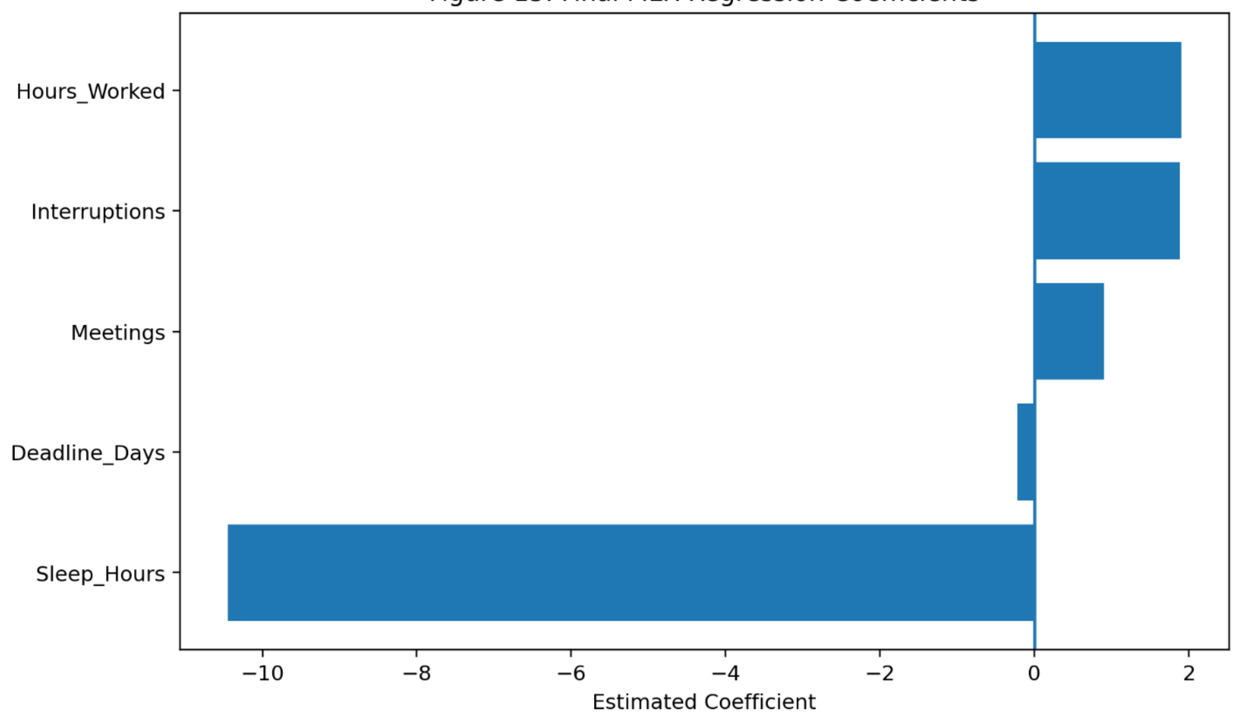
$$\widehat{Stress\_Level} = 94.78874 + 2.00258(Hours\_Worked) - 9.94568(Sleep\_Hours) + 0.12377(Bugs) - 0.20964(Deadline\_Days) + 0.97163(Meetings) + 2.04272(Interruptions)$$

Table 4.2: Final Multiple Linear Regression Coefficients

Predictor	Estimate	Std. Error	t-value	p-value
Intercept	**94.78874**	**4.89736**	**19.355**	$< 2 \times 10^{-16}$
Hours_Worked	**2.00258**	**0.24590**	**8.144**	4.63×10^{-15}
Sleep_Hours	**−9.94568**	**0.48516**	**−20.500**	$< 2 \times 10^{-16}$
Bugs	**0.12377**	**0.06031**	**2.052**	**0.0408**
Deadline_Days	**−0.20964**	**0.05071**	**−4.134**	3.42×10^{-5}
Meetings	**0.97163**	**0.13960**	**6.960**	1.35×10^{-11}
Interruptions	**2.04272**	**0.28678**	**7.123**	4.76×10^{-12}

All six predictors were statistically significant at the 5% significance level.

Figure 15. Final MLR Regression Coefficients



4.3.2 Interpretation of Regression Coefficients

Sleep_Hours had the largest absolute coefficient. Holding the other variables constant, an additional hour of sleep was associated with an estimated 9.95-point decrease in the predicted stress score.

Hours_Worked had a positive coefficient of 2.003, indicating that an additional hour worked was associated with an estimated 2.00-point increase in predicted stress.

Similarly, each additional interruption was associated with an estimated 2.04-point increase in stress. Meetings also had a positive association, with each additional meeting associated with approximately a 0.97-point increase in predicted stress.

The coefficient for Bugs was positive but comparatively small, while Deadline_Days showed a statistically significant negative association with stress.

These results describe statistical associations after controlling for the other variables in the model and should not be interpreted as evidence of direct causation.

4.3.3 Overall Model Fit

Table 4.3: Final MLR Model Fit

Measure	Value
Multiple R^2	0.6085
Adjusted R^2	0.6028
Residual Standard Error	17.87
F-statistic	106.5
Model p-value	$< 2.2 \times 10^{-16}$

The model explained approximately 60.85% of the variation in developer stress levels. After accounting for the number of predictors, the adjusted R^2 was 60.28%.

The overall F-test was statistically significant, indicating that the predictors collectively provide significant information for explaining variation in Stress_Level.

4.3.4 Multicollinearity Assessment

Table 4.4: Variance Inflation Factors

Predictor	VIF
Hours_Worked	1.019
Sleep_Hours	1.008
Bugs	1.018
Deadline_Days	1.004
Meetings	1.012
Interruptions	1.012

All VIF values are extremely close to 1. Therefore, there is no evidence of problematic multicollinearity among the predictors included in the final MLR model.

4.3.5 Final MLR Diagnostic Analysis

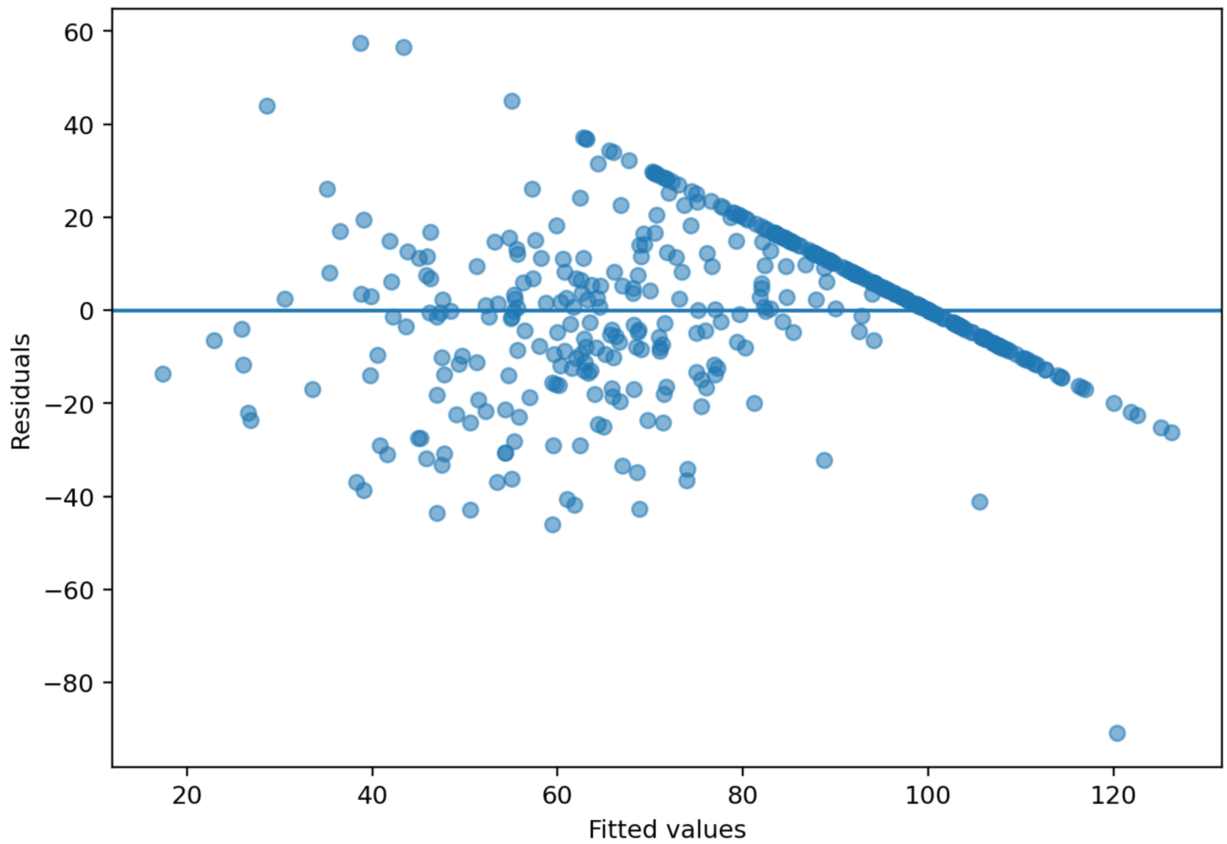


Figure 4.4: Residuals vs Fitted Values

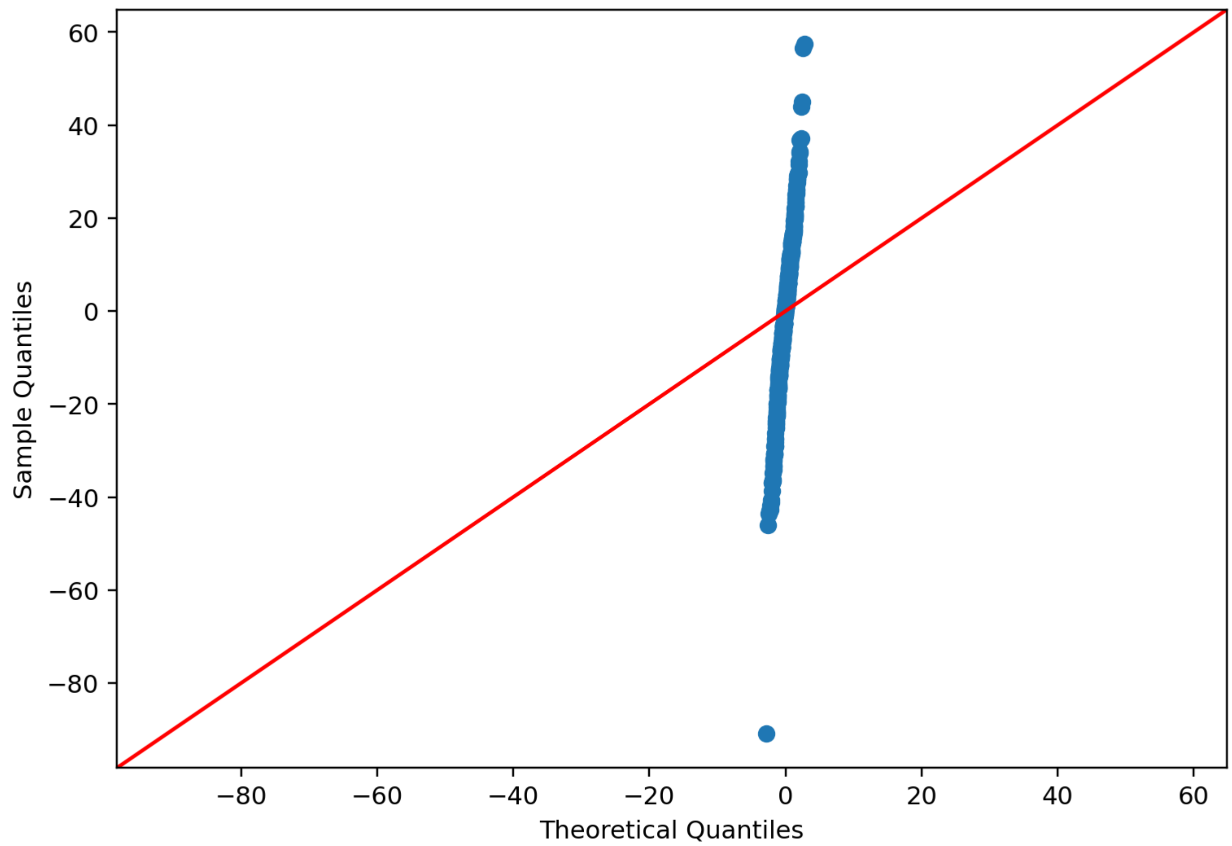


Figure 4.5: Normal Q-Q Plot

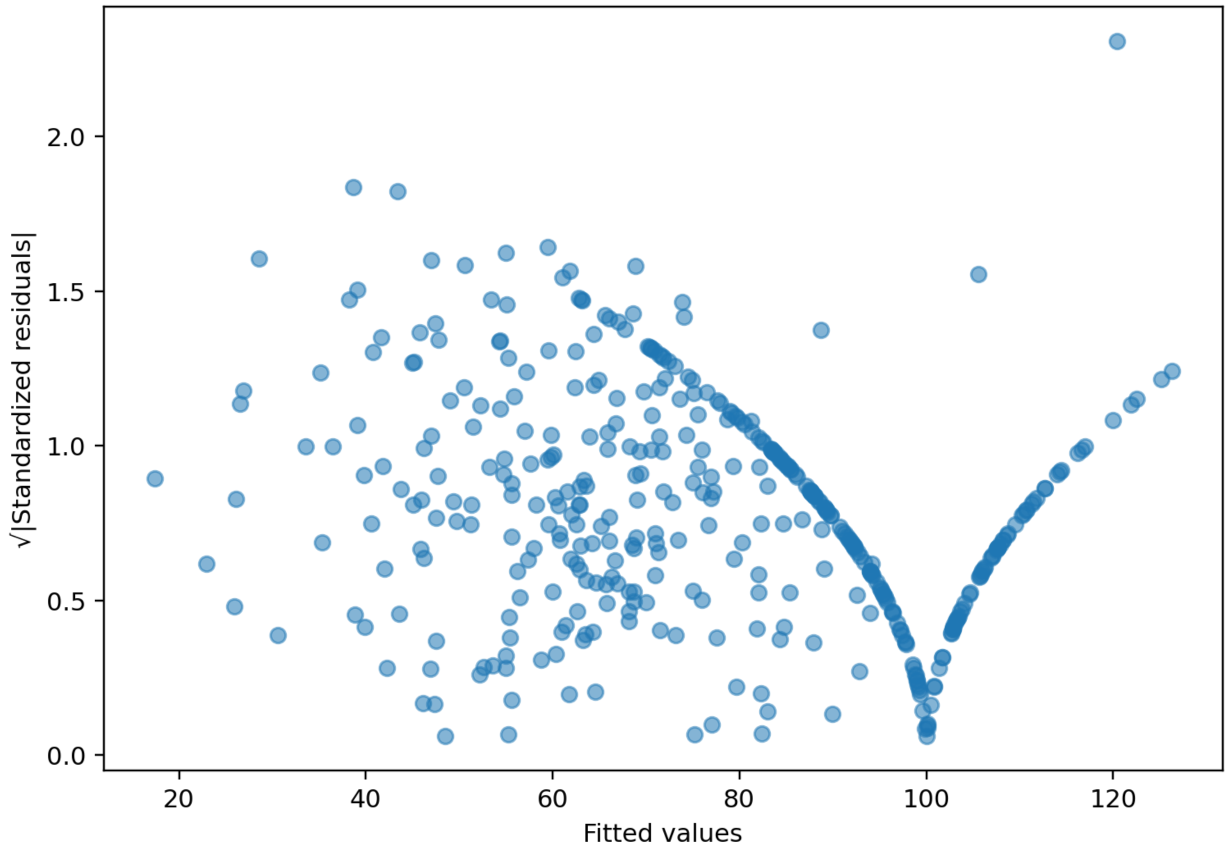


Figure 4.6: Scale-Location Plot

The residual diagnostic plots were examined to assess the assumptions of the linear regression model. The residuals versus fitted plot indicates a systematic pattern rather than a completely random distribution, particularly at higher fitted values. The Q-Q plot also indicates departures from normality, while the Scale-Location plot suggests non-constant residual variability.

4.3.6 MLR Predictive Performance

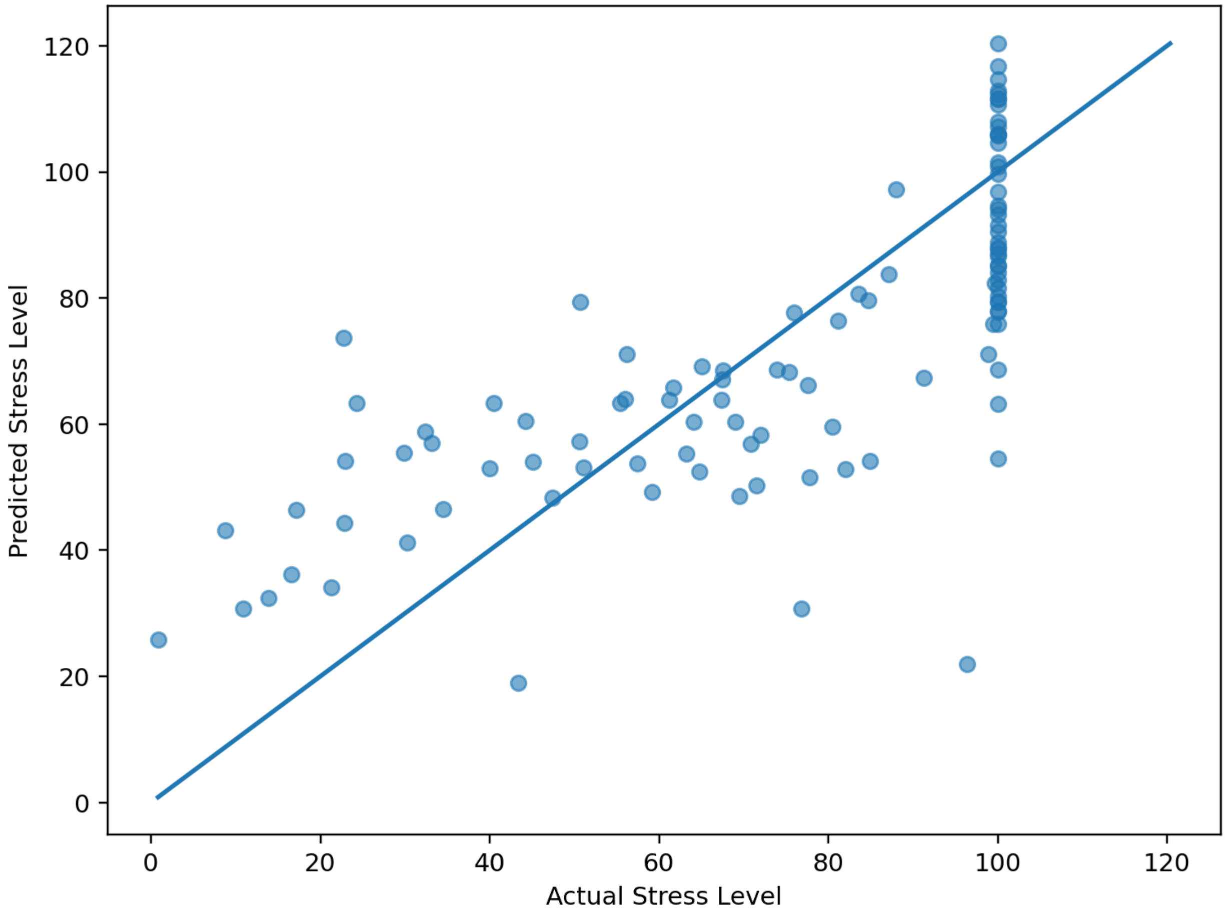


Figure 4.7: Actual vs Predicted Stress Level

The actual-versus-predicted plot shows a positive relationship between observed and predicted stress levels, indicating that the model captures a substantial part of the variation in stress.

Table 4.5: Test-Set Predictive Performance

Metric	Value
R^2	**0.5885**
RMSE	**16.48**
MAE	**12.85**

The test-set R^2 of 0.5885 is reasonably close to the training R^2 of 0.6085. This indicates that the model retains a substantial proportion of its explanatory ability when applied to unseen observations.

The RMSE of 16.48 indicates the typical magnitude of prediction error on the stress scale, while the MAE of 12.85 indicates the average absolute prediction error.

4.4 Ordinal Logistic Regression

4.4.1 Construction of Stress Categories

To provide a complementary analysis, the continuous stress score was transformed into an ordered categorical response consisting of:

Low < Moderate < High

The categories were used specifically for the ordinal logistic regression analysis, while the original continuous Stress_Level remained the response for MLR.

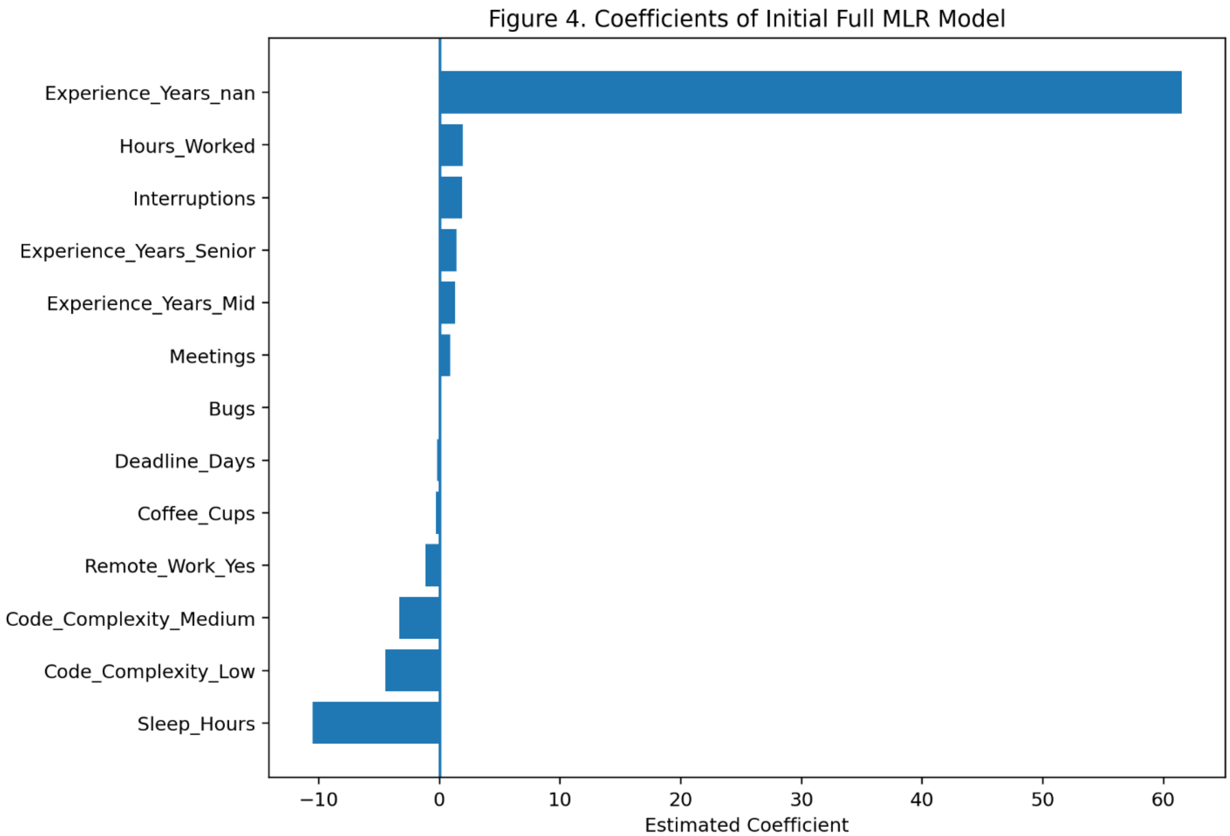


Figure 4.8: Distribution of Training Stress Categories

The category distribution shows that the High stress category contains the largest number of observations, followed by the Low and Moderate categories. This imbalance is important when interpreting the classification results, particularly the performance for the Moderate category.

4.4.2 Final Ordinal Logistic Regression Model

The ordinal logistic regression model used five predictors:

- Hours_Worked
- Sleep_Hours
- Deadline_Days
- Meetings
- Interruptions

Bugs was removed during the stepwise model-selection procedure.

The final ordinal model therefore provides a more parsimonious representation of the factors associated with movement toward higher stress categories.

4.4.3 Odds Ratio Analysis

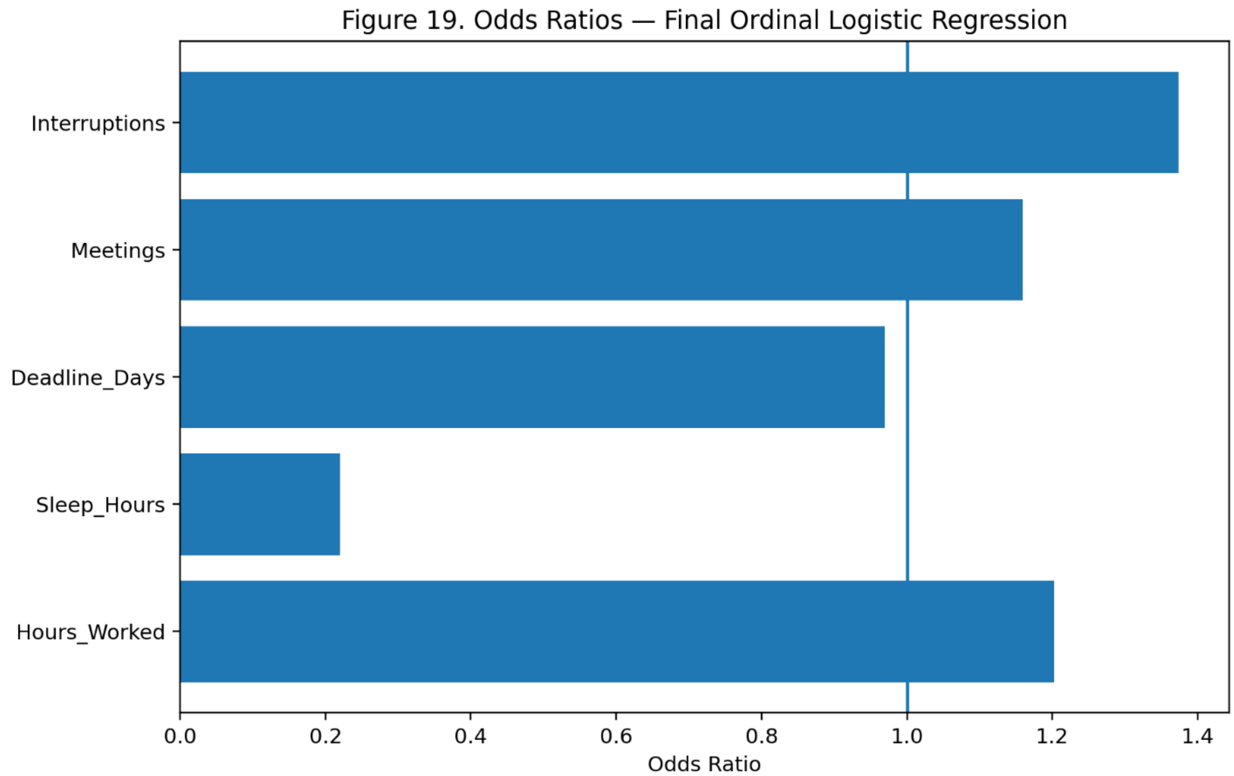


Figure 4.9: Odds Ratios for the Final Ordinal Logistic Model

The odds-ratio results show that Hours_Worked, Meetings, and Interruptions have odds ratios greater than 1, indicating higher odds of belonging to a higher stress category as these variables increase.

In contrast, Sleep_Hours has an odds ratio substantially below 1, indicating that greater sleep duration is associated with lower odds of being in a higher stress category.

Sleep_Hours therefore represents one of the most influential variables in the ordinal analysis. Interruptions also shows a relatively strong positive association with higher stress categories.

4.4.4 Proportional-Odds Assumption

The proportional-odds assumption was assessed using the Brant test.

The overall test was statistically significant:

$$\chi^2 = 30.81,\ \mathrm{df} = 5,\ \mathrm{p} < 0.001$$

Therefore, the proportional-odds assumption was not fully satisfied.

At the individual predictor level, Sleep_Hours was the main variable showing evidence of violating the assumption. This indicates that the effect of sleep duration is not constant across all cumulative stress-category thresholds.

To investigate this further, threshold-specific binary logistic models were examined. The estimated effect of Sleep_Hours differed between the thresholds, supporting the conclusion that its relationship with stress severity changes across the category boundaries.

This finding should be acknowledged as an important limitation of the standard ordinal logistic model.

4.4.5 Threshold-Specific Analysis

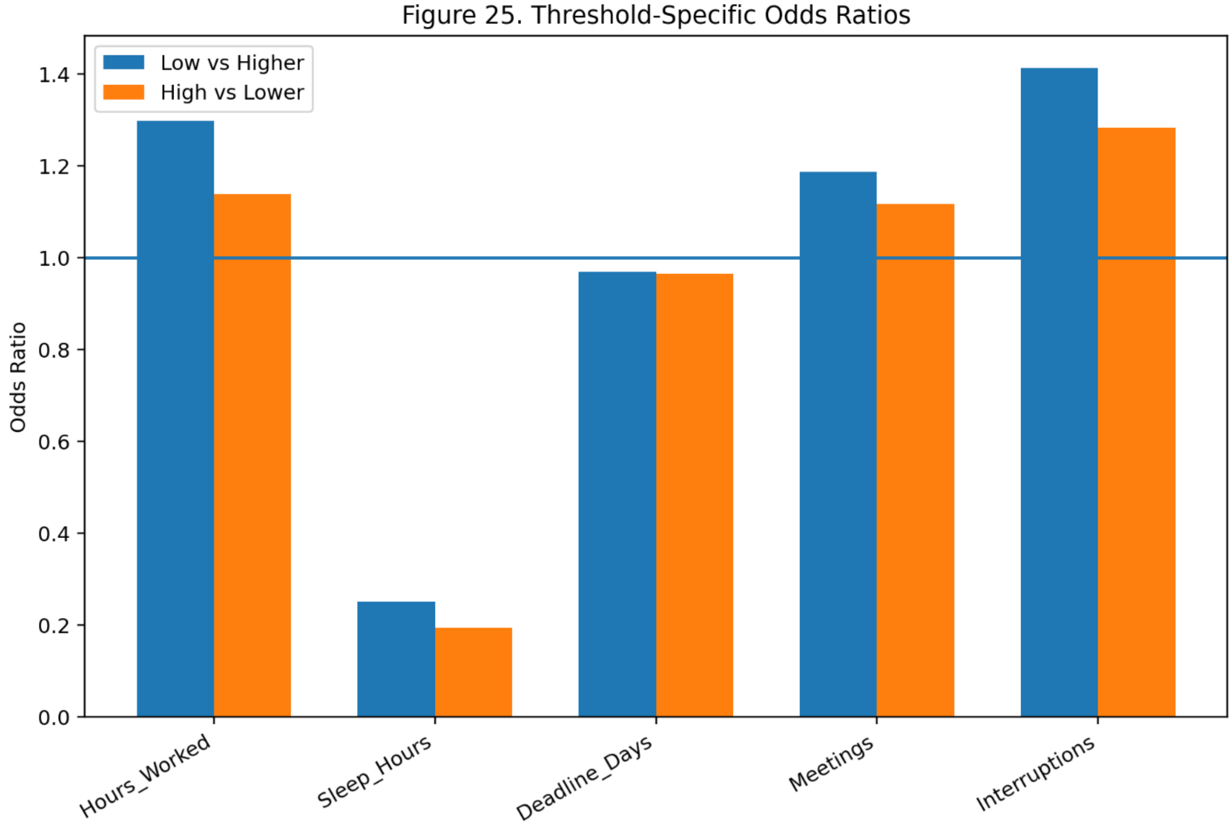


Figure 4.11: Threshold-Specific Odds Ratios

The threshold-specific analysis demonstrates that the effects of the predictors can differ depending on the stress threshold being considered. In particular, the effect of Sleep_Hours changes between the cumulative comparisons.

This provides additional evidence explaining the violation of the proportional-odds assumption identified by the Brant test.

4.4.6 Ordinal Logistic Classification Performance

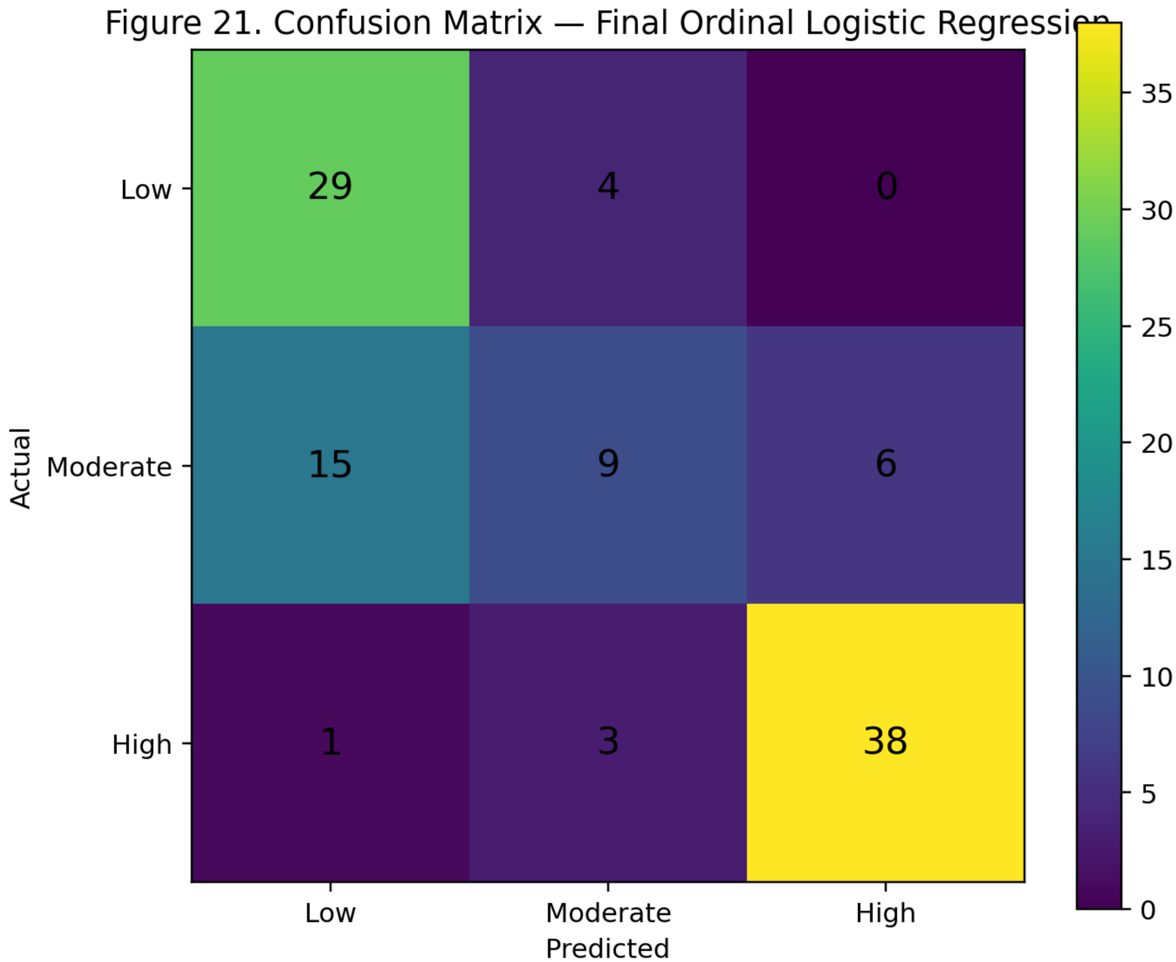


Figure 4.12: Confusion Matrix of the Final Ordinal Logistic Model

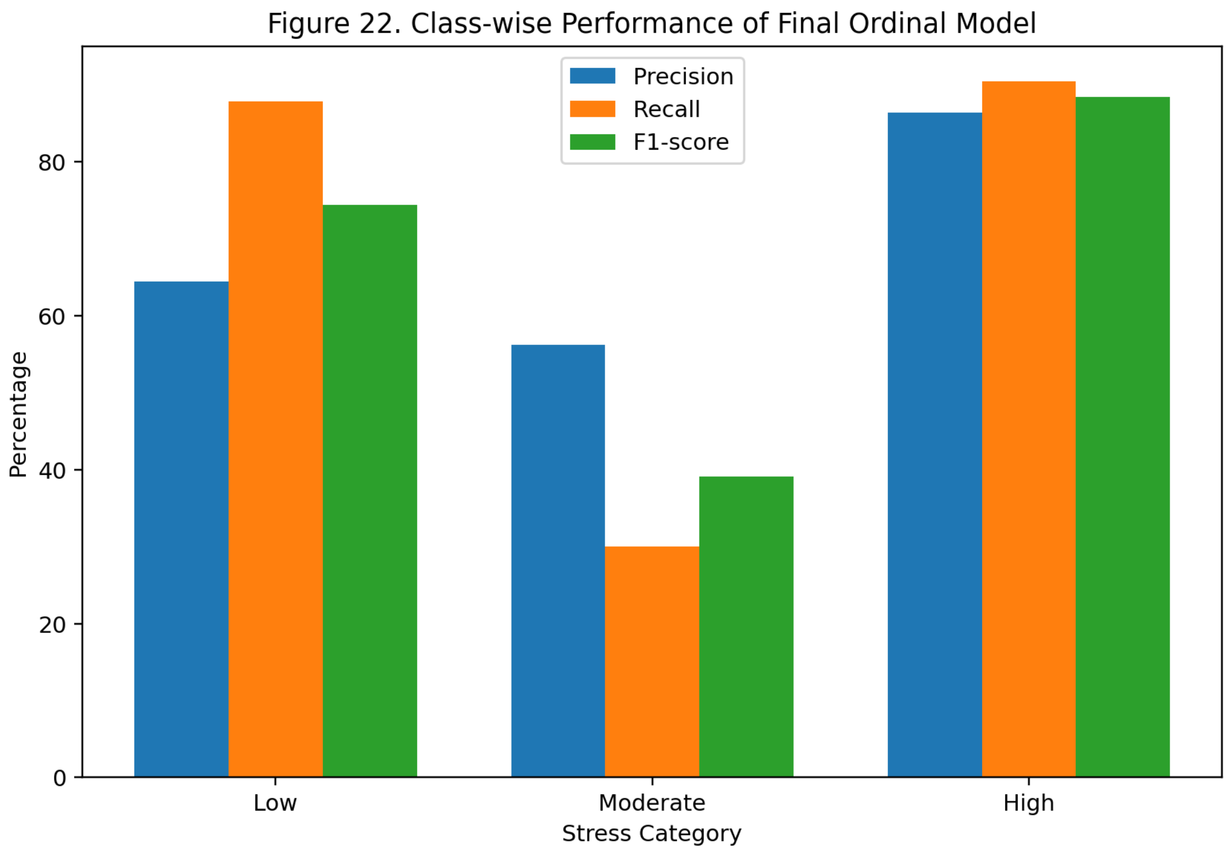


Figure 4.13: Class-wise Classification Performance

The confusion matrix shows that the model correctly classified:

- 29 Low observations,
- 9 Moderate observations,
- 38 High observations.

The largest number of misclassifications occurs for the Moderate category. This indicates that Moderate stress is more difficult for the model to distinguish from Low and High stress than the two extreme categories.

Based directly on the displayed confusion matrix, the classification performance is:

Table 4.6: Classification Performance

Stress Category	Precision (%)	Recall (%)	F1-score (%)
Low	**64.44**	**87.88**	**74.36**
Moderate	**56.25**	**30.00**	**39.13**
High	**86.36**	**90.48**	**88.37**
Overall Accuracy			**72.38%**

The model performs particularly well in identifying High-stress developers, with a recall of approximately 90.48%. Performance for the Moderate category is considerably weaker, with a recall of only 30%.

Overall, the displayed confusion matrix gives an accuracy of approximately 72.38%. This should be the value reported if this confusion matrix is the final test-set output.

4.5 Comparison of MLR and Ordinal Logistic Regression

The two approaches provide complementary information about developer stress.

Table 4.7: Comparison of Final Models

Aspect	Multiple Linear Regression	Ordinal Logistic Regression
Response	**Continuous Stress_Level**	**Low, Moderate, High**
Main purpose	**Predict stress score**	**Predict stress category**
Final predictors	**6**	**5**
R^2	**0.6085**	—
Adjusted R^2	**0.6028**	—
Test R^2	**0.5885**	—
Test RMSE	**16.48**	—
Test MAE	**12.85**	—
Classification Accuracy	—	**72.38%**
Main diagnostic issue	**Non-constant variance / residual departures**	**Proportional-odds violation**
Strongest factor	**Sleep_Hours**	**Sleep_Hours**

The MLR model is useful when the objective is to estimate a numerical stress score, whereas the ordinal logistic model is useful when the objective is to identify the severity category of stress.

The two models consistently identify Sleep_Hours as an important factor. Work-related factors such as Hours_Worked, Meetings, and Interruptions also contribute positively to predicted stress.

4.6 Discussion of Findings

The results indicate that both personal recovery and work intensity are important in explaining developer stress.

The strongest finding across the analyses is the relationship between sleep and stress. The MLR model shows a substantial negative coefficient for Sleep_Hours, while the ordinal model produces an odds ratio well below 1. Together, these results indicate that lower sleep duration is strongly associated with higher developer stress.

Work intensity also plays an important role. Hours_Worked has a positive coefficient in the MLR model and an odds ratio above 1 in the ordinal model. Similarly, Meetings and Interruptions are positively associated with stress.

The ordinal analysis adds another perspective by showing that the model is considerably better at distinguishing Low and High stress than Moderate stress. This may be partly related to the distribution of observations across the three categories.

Overall, the findings suggest that statistical modeling can provide useful information about developer stress, but the results should be interpreted within the limitations identified through model diagnostics and the nature of the dataset.

4.7 Summary of Chapter

The exploratory analysis identified important patterns in the developer stress data, particularly the negative association between sleep duration and stress and the positive relationships involving working hours, meetings, and interruptions.

The final MLR model explained approximately 60.85% of the variation in stress levels and achieved a test-set R^2 of 0.5885, with RMSE of 16.48 and MAE of 12.85.

The ordinal logistic model provided a complementary classification perspective. It retained five predictors and achieved an overall classification accuracy of approximately 72.38% based on the displayed confusion matrix. However, the proportional-odds assumption was violated, mainly due to Sleep_Hours, and this limitation should be considered when interpreting the ordinal model.

Overall, the results demonstrate that sleep, working hours, meetings, and interruptions are important factors associated with software developer stress, while the two modeling approaches provide complementary continuous and categorical perspectives.

05.Conclusion and Recommendations

5.1 Introduction

This chapter summarizes the main findings of the study on predicting software developer stress levels using multiple linear regression and ordinal logistic regression. The conclusions drawn from both modeling approaches are discussed, followed by practical recommendations, limitations, and directions for future research.

5.2 Summary of the Study

This study investigated factors associated with software developer stress using statistical modeling techniques. The analysis considered workload, sleep, technical, and work-environment characteristics as potential predictors of stress.

Two complementary modeling approaches were used. Multiple linear regression was applied to the continuous Stress_Level score, while ordinal logistic regression was used to classify developers into ordered Low, Moderate, and High stress categories. Exploratory data analysis, variable selection, multicollinearity assessment, model diagnostics, and predictive evaluation were conducted to assess the suitability and performance of the models.

The final regression model considered the following six predictors:

- Hours_Worked
- Sleep_Hours
- Bugs
- Deadline_Days
- Meetings
- Interruptions

The categorical variables Experience_Years, Code_Complexity, and Remote_Work were not retained in the final model.

5.3 Main Findings from Multiple Linear Regression

The final multiple linear regression model included Hours_Worked, Sleep_Hours, Bugs, Deadline_Days, Meetings, and Interruptions. All six predictors were statistically significant at the 5% significance level.

Sleep_Hours showed the strongest negative association with stress. In contrast, Hours_Worked, Bugs, Meetings, and Interruptions showed positive associations with the predicted stress level. Deadline_Days showed a negative association in the fitted model.

The final MLR model achieved an R^2 of **0.6085** and an adjusted R^2 of **0.6028**, explaining approximately 60% of the variation in developer stress levels. On the test data, the model achieved an R^2 of **0.5885**, with an RMSE of **16.48** and MAE of **12.85**.

5.4 Main Findings from Ordinal Logistic Regression

The ordinal logistic regression analysis provided a complementary perspective by modeling stress as three ordered categories: Low, Moderate, and High. After stepwise variable selection, Hours_Worked, Sleep_Hours, Deadline_Days, Meetings, and Interruptions were retained, while Bugs was removed.

The results were generally consistent with the MLR analysis. Hours_Worked, Meetings, and Interruptions were associated with higher odds of belonging to a higher stress category, while Sleep_Hours was associated with lower odds of higher stress. This consistency strengthens the evidence that these factors are important indicators of developer stress.

The ordinal model achieved an overall test classification accuracy of approximately **72.38%** based on the final confusion matrix. The model performed particularly well in identifying Low and High stress observations, while classification of the Moderate category was more difficult.

5.5 Comparison of the Two Models

The two statistical approaches provided complementary information. Multiple linear regression was useful for predicting the magnitude of the continuous stress score, whereas ordinal logistic regression provided information about the likelihood of developers belonging to higher stress categories.

Both approaches identified Sleep_Hours, Hours_Worked, Meetings, and Interruptions as important factors. However, the ordinal model reduced the number of predictors by removing Bugs, resulting in a more parsimonious classification model.

Therefore, the MLR model is more appropriate when an estimated numerical stress score is required, while the ordinal logistic model is useful when the objective is to identify the stress severity category.

5.6 Practical Recommendations

5.6.1 Encourage Adequate Sleep

Since sleep duration showed a strong negative relationship with stress in both modeling approaches, organizations should promote healthy work schedules that provide developers with sufficient opportunities for rest and recovery.

5.6.2 Manage Working Hours

The positive association between working hours and stress suggests that excessive working hours should be monitored. Organizations should aim to distribute workloads appropriately and avoid prolonged periods of excessive work.

5.6.3 Reduce Unnecessary Interruptions

Interruptions were positively associated with both continuous stress and higher stress categories. Development teams could therefore introduce protected periods of uninterrupted work and reduce unnecessary notifications and ad-hoc interruptions.

5.6.4 Optimize Meetings

Since meetings showed a positive association with stress, organizations should evaluate the necessity and frequency of meetings and encourage shorter, focused meetings where appropriate.

5.6.5 Use Statistical Models as Support Tools

The developed models could potentially be used as analytical support tools to identify patterns associated with elevated developer stress. However, predictions should be used to support employee well-being rather than for employee surveillance or performance evaluation.

5.7 Limitations

5.7.1 Simulated Dataset

The study is based on a developer stress simulation dataset. Therefore, the findings may not fully represent stress patterns among software developers in real-world organizations.

5.7.2 Regression Assumptions

The MLR diagnostic analysis indicated departures from some regression assumptions. Therefore, the model results should be interpreted with appropriate caution.

5.7.3 Proportional-Odds Assumption

The ordinal logistic regression model did not fully satisfy the proportional-odds assumption. The Brant test indicated a statistically significant overall violation, with Sleep_Hours being the main predictor associated with this violation. Threshold-specific analyses suggested that the effect of sleep duration differs across stress-category thresholds. Consequently, the ordinal logistic results should be interpreted with caution.

5.7.4 Moderate Stress Classification

The ordinal model showed weaker predictive performance for the Moderate stress category compared with the Low and High categories. This indicates that distinguishing intermediate stress levels may be more difficult using the available predictors.

5.7.5 Generalizability

The findings should not be generalized to all software developers because the dataset may not capture the diversity of real-world development environments and individual circumstances.

5.8 Future Work

1. Use real-world developer data collected through surveys or organizational studies.
2. Include additional predictors, such as job satisfaction, work-life balance, organizational support, team relationships, management style, and burnout indicators.
3. Investigate nonlinear relationships because the effects of factors such as working hours and sleep may not always be linear.
4. Investigate interaction effects, such as whether the effect of working hours depends on sleep duration.

5. Explore alternative ordinal models that do not require the proportional-odds assumption, since the Brant test indicated a violation.
6. Use larger and more balanced datasets to improve classification of the Moderate stress category.
7. Perform external validation using an independent dataset to determine how well the models generalize to new developer populations.

5.9 Final Conclusion

This study demonstrated the usefulness of statistical modeling for investigating and predicting software developer stress. Multiple linear regression successfully modeled the continuous stress score, explaining approximately 60.85% of its variation, while the ordinal logistic regression provided a complementary approach for predicting ordered stress severity categories.

Across both approaches, Sleep_Hours, Hours_Worked, Meetings, and Interruptions emerged as important factors associated with developer stress. In particular, greater sleep duration was consistently associated with lower stress, whereas greater working hours, meetings, and interruptions were associated with increased stress.

Although the models provided useful predictive information, their limitations must be considered. The MLR diagnostics indicated some departures from model assumptions, while the ordinal logistic model did not fully satisfy the proportional-odds assumption. Therefore, the models should be viewed as statistical tools for identifying patterns and supporting further investigation rather than as definitive causal or diagnostic instruments.

Overall, the study provides evidence that statistical modeling can offer valuable insights into the factors associated with software developer stress and can support data-driven approaches to improving developer well-being.